

Generalized Pareto distribution: $P(Y > y) = \left(1 - \frac{y}{\tau}\right)^\lambda$ for $0 \leq y < \tau$, $\lambda > 0$

For $0 \leq y < \tau$: $P(Y > y) = \left(1 - \frac{y}{\tau}\right)^\lambda \Rightarrow F(y) = P(Y \leq y) = 1 - P(Y > y)$

thus, $F(y) = 1 - \left[1 - \left(\frac{y}{\tau}\right)^\lambda\right]$. Hence, the density is

$$f_Y(y; \tau, \lambda) = \lambda \left[1 - \frac{y}{\tau}\right]^{\lambda-1} \cdot \frac{1}{\tau} = \frac{\lambda}{\tau} \left[1 - \frac{y}{\tau}\right]^{\lambda-1}$$

$$\Rightarrow \log\{f_Y(y; \tau, \lambda)\} = \log\{\lambda\} - \log\{\tau\} + (\lambda-1) \log\left\{1 - \frac{y}{\tau}\right\} = \ell(y; \tau)$$

So, we have $u(y; \tau) = \frac{\partial \ell}{\partial \tau} = -\frac{1}{\tau} + (\lambda-1) \frac{\partial}{\partial \tau} \log\left\{1 - \frac{y}{\tau}\right\}$

$$\Rightarrow u(y; \tau) = -\frac{1}{\tau} + (\lambda-1) \cdot \frac{y/\tau^2}{1 - y/\tau}$$

we need to integrate it and equate to 1

1st Barlett identity: $E[u(Y; \tau)] = 0$. Using the fact that $\int_Y f_Y(y; \tau) dy = 1$

$$\Rightarrow 0 = \frac{\partial}{\partial \tau} \int f_Y(y; \tau) dy = \int \frac{\partial}{\partial \tau} f_Y(y; \tau) dy = \int \frac{\partial \log f}{\partial \tau} f_Y(y; \tau) dy$$

$$\Rightarrow 0 = E\left[\frac{\partial \log f_Y(Y; \tau)}{\partial \tau}\right] = E[u(Y; \tau)] \text{ as long as}$$

we may differentiate under the integral sign and $\lambda > 0$

2nd Barlett identity: $E\left[\left(\frac{\partial \log f}{\partial \tau}\right)^2\right] = E[u(Y; \tau)^2] = -E\left[\frac{\partial^2 \log f}{\partial \tau^2}\right]$

Note that this requires both expectations to be finite!

Trouble is at endpoint $y = \tau$ but we may look at the behaviour near $y = \tau$:

Let $t = \tau - y \Rightarrow 1 - \frac{y}{\tau} = 1 - \frac{\tau - t}{\tau} = \frac{t}{\tau}$ (near $y = \tau \Rightarrow t \approx 0$)

So, the density behaves like $f_Y(y; \tau, \lambda) = \frac{\lambda}{\tau} \left(1 - \frac{y}{\tau}\right)^{\lambda-1}$ hence,

$$\lim_{y \rightarrow \tau^-} \frac{f_Y(y; \tau, \lambda)}{t^{\lambda-1}} \rightarrow c > 0 \text{ i.e. } f_Y(y; \tau, \lambda) \approx c t^{\lambda-1} \text{ for } c > 0.$$

And thus, $u(y; \tau) = -\frac{1}{\tau} + (\lambda-1) \frac{y/\tau^2}{1 - y/\tau} \approx -\frac{1}{\tau} + (\lambda-1) \frac{1/\tau}{t/\tau} \approx \frac{\lambda-1}{t}$

$$\Rightarrow u(y; \tau)^2 \sim \frac{(\lambda-1)^2}{t^2}$$

Therefore, near the endpoint,

$$E[u(Y; \tau)^2] = \int u(y; \tau)^2 f_Y(y; \tau, \lambda) dy \text{ and it behaves like}$$

$$u(y; \tau)^2 f_Y(y; \tau, \lambda) \approx t^{\lambda-1} t^{-2} = t^{\lambda-3} \text{ where this integral}$$

converges iff $\lambda-3 > -1 \Leftrightarrow \lambda > 2$

• If $\lambda > 2 \Rightarrow E[u(Y; \tau)^2] < \infty$ and 2nd identity holds!

• If $0 < 2 \leq \lambda \Rightarrow E[u(Y; \tau)^2] = \infty$ and 2nd identity fails!

Now, let $Y_1, \dots, Y_n$ be i.i.d. $\sim$ Generalized Pareto, and
 $M_n := \max\{Y_1, \dots, Y_n\}$. We want the limit law of
 $W_n := n^{1/\lambda}(\gamma - M_n)$:

For $w \geq 0$: $P(W_n > w) = P(\gamma - M_n > wn^{-1/\lambda}) = P(M_n < \gamma - wn^{-1/\lambda})$

Since Y_i 's are indep. $\Rightarrow P(M_n \leq x) = [F_X(x)]^n$

Using F_X and the point $x = \gamma - wn^{-1/\lambda}$, we note the fact that

$$F_X(\gamma - wn^{-1/\lambda}) = 1 - \left(\frac{wn^{-1/\lambda}}{\gamma}\right)^\lambda = 1 - \frac{(w/\gamma)^\lambda}{n} \quad \text{Since: } F_X(\gamma - z) = 1 - \left[1 - \frac{\gamma - z}{\gamma}\right]^\lambda = 1 - \left(\frac{z}{\gamma}\right)^\lambda$$

Hence, $P(W_n > w) = \left[1 - \frac{(w/\gamma)^\lambda}{n}\right]^n \xrightarrow{n \rightarrow \infty} \exp\left\{-\left(\frac{w}{\gamma}\right)^\lambda\right\}$. This yields

$W_n \rightarrow W$ where the cdf of W : $P(W \leq w) = 1 - \exp\left\{-\left(\frac{w}{\gamma}\right)^\lambda\right\}$.
for $w \geq 0$.

Thus, we may deduce that the "natural" estimator $\gamma - W_n/n^{1/\lambda}$ (based on the maximum) converges at rate $n^{-1/\lambda}$:

$$n^{1/\lambda}(\gamma - M_n) \xrightarrow{d} W !!! \quad \leftarrow \text{when it exists and barlett's id. hold}$$

By contrast, a regular MLE $\hat{\tau}$ would typically go as
 $\sqrt{n}(\hat{\tau} - \gamma) \xrightarrow{d} \mathcal{N}(0, I(\gamma)^{-1})$ (converges at usual rate $\sqrt{n}$)

If $\lambda \leq 2 \Rightarrow \frac{1}{\lambda} \geq \frac{1}{2}$ so $n^{1/\lambda} \gg \sqrt{n}$ as $n \rightarrow \infty$

This means that $|\gamma - M_n|$ "shrinks" faster than $|\hat{\tau} - \gamma|$

// Convergence to the limiting distribution based on the maximum happens faster than the usual $\sqrt{n}$ rate of MLE //